

Key Points from the PDF (Concepts that motivated our project)

Cyberspace

"A global domain within the information environment consisting of the interdependent network of information technology infrastructures, including the Internet, telecommunications networks, computer systems, and embedded processors and controllers."

-- A Definition of Cyberspace
(Page 2)

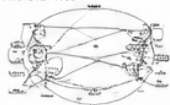
- Cyberspace includes interconnected networks, systems, and devices.
- Everything is connected and interdependent.
- This visibility is essential to protect information and critical infrastructure.

Life in a Networked World

- **Rapid Development in Information Technology**
 - Speed of Microprocessor chips doubles every 12-18 months
 - Storage Density doubles every 12 months
 - Bandwidth is doubling every 12 months
 - Price keeps dropping making technology affordable & pervasive

The Old "Net"

The New "Net"



(Page 3)

- Rapid growth of networks, devices and bandwidth.
- Low cost and high connectivity make technology widely accessible.
- The new network monitors and controls critical infrastructure. Its integrity and availability is critical for economy, public safety and national security.

Cyber What? Defining Cyber

- Cyberspace is the connected Internet Ecosystem
- Trends Exposing critical infrastructure to increased risk:
 - Interconnectedness of Sectors
 - Proliferation of exposure points
 - Concentration of Assets
- Cyber Intrusions and Attacks have increased dramatically over the last decade, exposing sensitive information, disrupting critical operations, and imposing high costs.
- Cyber Security is protecting our cyber space from attack, damage, misuse and economic espionage.

(Page 5)

- Increased interconnection and more exposure points lead to higher risk.
- Attacks can disrupt critical operations and cause large economic impact.
- Cyber security aims to protect systems from attacks, misuse and damage.

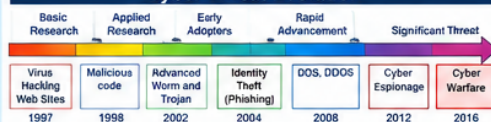
Cyber Security Challenges

- Cyberspace has inherent vulnerabilities that cannot be removed
- Innumerable entry points to internet.
- Assigning attribution is difficult.
- Defense techniques largely protect individual systems rather than critical operations.
- Attack technology outpacing defense technology
- Nation states, non-state actors, and individuals can all launch attacks (peer level).

(Page 8)

- There are many entry points and inherent vulnerabilities.
- Attacks are getting more sophisticated (nation-states, non-state actors, individuals).
- Traditional defenses are not enough.
- Need continuous monitoring, analysis and advanced techniques.

Cyber Threat Evolution



Cyber exploitation activity becoming More Sophisticated, More Targeted and More Serious

(Page 8)

- Cyber threats have evolved from simple viruses to advanced malware, botnets, cyber espionage and cyber warfare.
- Attacks are becoming more targeted, sophisticated and serious over time.

How it Motivated and Influenced NeuroShield (What we implemented / Our Approach)



Inspired a network-centric approach

- We model the network as a dynamic graph (Source IP → Destination IP) to understand relationships between devices.
- Monitor overall network behavior instead of isolated events.
- Helps detect threats across interconnected systems.



Motivated real-time and scalable monitoring

- We process network flow data in 30-second windows over large datasets (CIC-IDS2017).
- Built a scalable pipeline (FastAPI backend + React SOC).
- Focus on real-time detection to protect critical systems from growing cyber threats.



Motivated threat-focused detection

- We detect abnormal network behavior using GraphSAGE + LSTM to identify intrusions and potential future attacks (T+1, T+3).
- Helps prevent disruption of critical services.
- Provides attack probability and MITRE ATT&CK stage to understand attacker intent (e.g., initial access, execution, etc.).



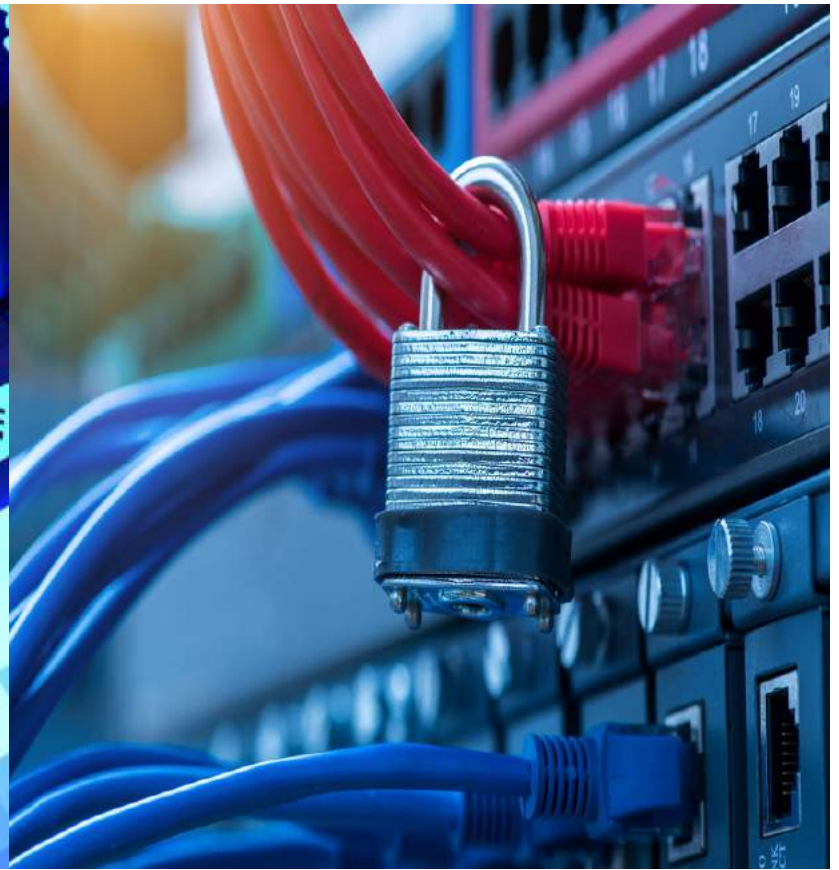
Motivated AI/ML and graph-based approach

- We use 25 behavioral features from network flows.
- GraphSAGE captures network relationships and LSTM models temporal evolution.
- Helps detect sophisticated and evolving attack patterns beyond simple signature-based methods.



Motivated future attack prediction and proactive defense

- We not only detect current attacks but also predict future network state and attack probability (World Model, T+1, T+3).
- Helps in early warning and proactive security response.
- Supports long-term cybersecurity for critical networks.



Cyber Security

by

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Member, NITI Aayog

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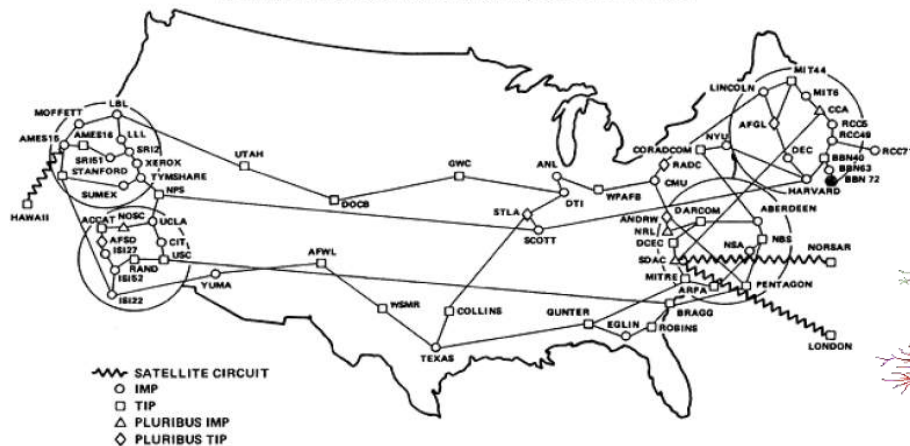
Life in a Networked World

- **Rapid Development in Information Technology**

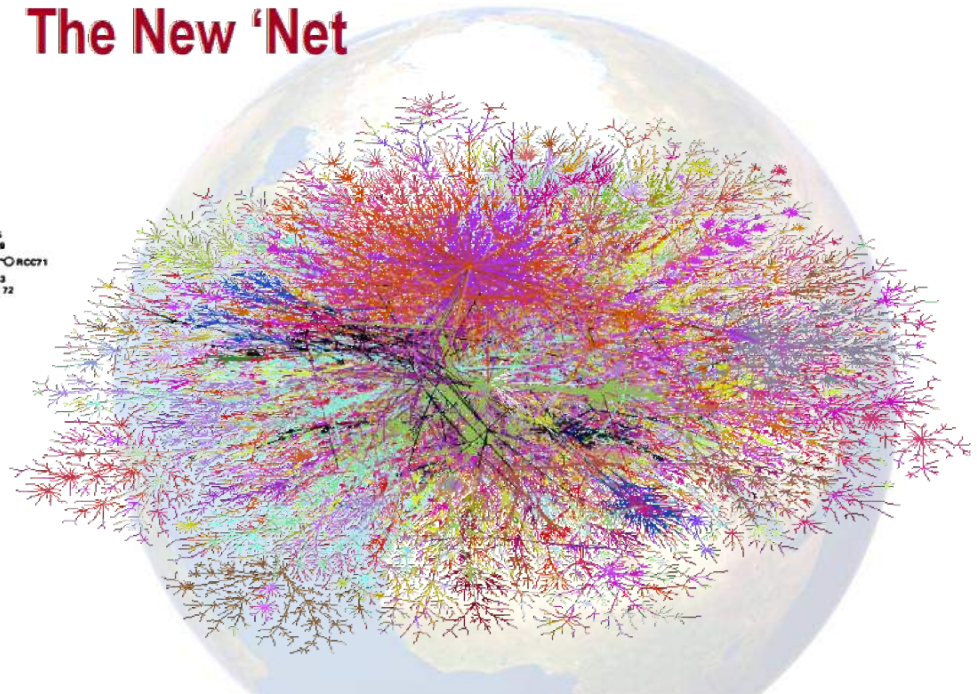
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The Old 'Net

ARPANET GEOGRAPHIC MAP, OCTOBER 1980



The New 'Net



The New “Net” monitors & controls critical Infrastructure. Its integrity & availability is critical for economy, public safety, & national security

The Internet In India by 2020



730

Million Internet
Users

75%

New users from
Rural Areas

75%

New users to consume data
in vernacular languages

83%

CAGR mobile video
content growth

175

Million on-line
shoppers

70%

E-commerce transactions
will be via mobile

50%

Travel transactions
will be online

Indian Economy Marching the e-way

Cyber What? Defining Cyber

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- Cyber Intrusions and Attacks have increased dramatically over the last decade, exposing sensitive personal and business information, disrupting critical operations, and imposing high costs on the economy
- Cyber Security is protecting our cyber space (critical infrastructure) from attack, damage, misuse and economic espionage



Food & Agriculture



Commercial Facilities



Dams



Energy



Information Technology



Postal & Shipping



Banking & Finance



Communication



Defence Industrial Base



Government Facilities



National Monuments & Icons



Transportation Systems



Chemical



Critical Manufacturing



Emergency Services



Healthcare & Public Health



Nuclear Reactors, Materials &
Wastes

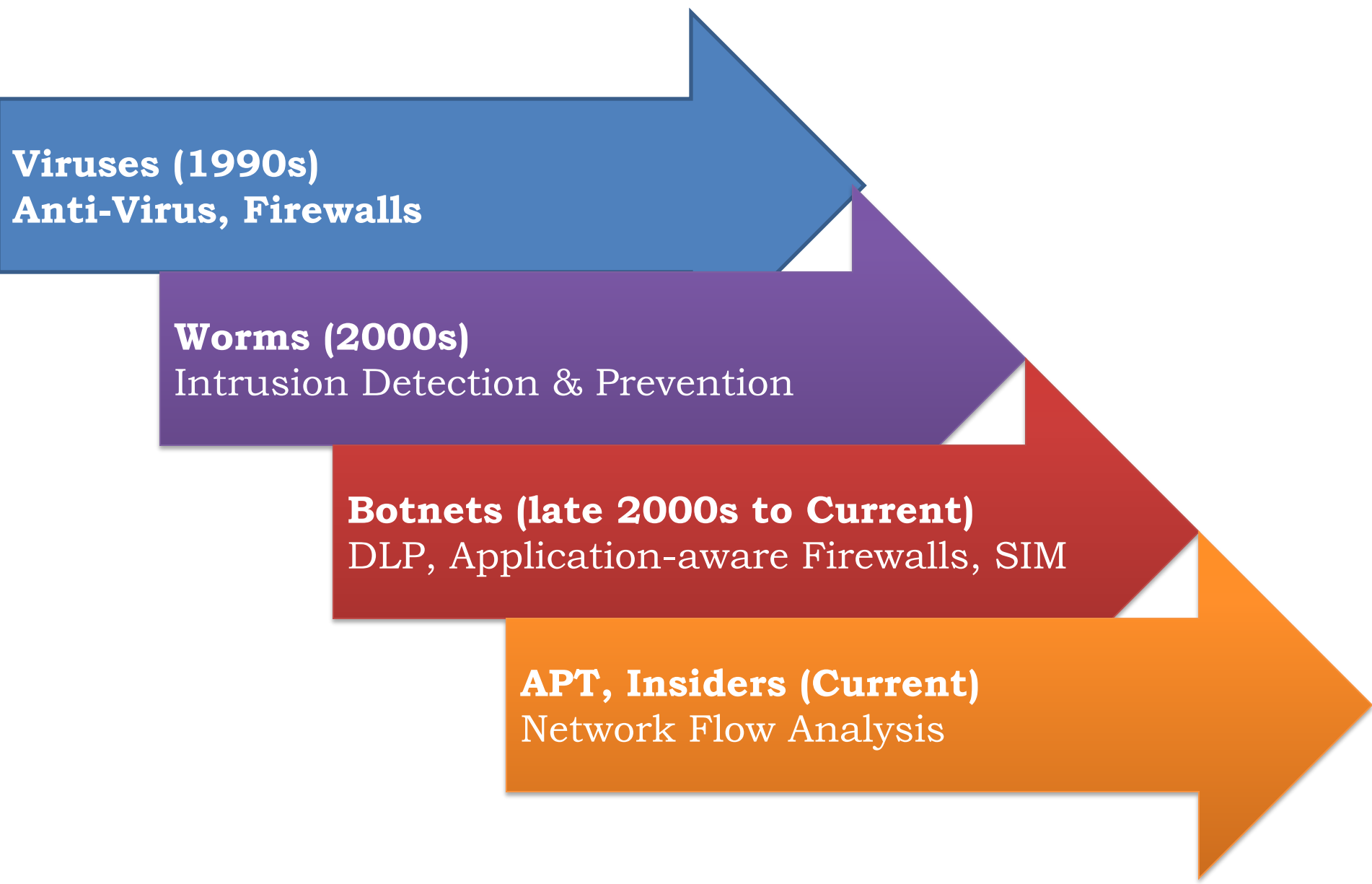


Water

Cyber Security Challenges

- Cyberspace has inherent vulnerabilities that cannot be removed
- Innumerable entry points to internet.
- Assigning attribution: Internet technology makes it relatively easy to misdirect attribution to other parties
- Computer Network Defense techniques, tactics and practices largely protect individual systems and networks rather than critical operations (missions)
- Attack technology outpacing defense technology
- Nation states, non-state actors, and individuals are at a peer level, all capable of waging attacks

Evolution Of Cyber Security



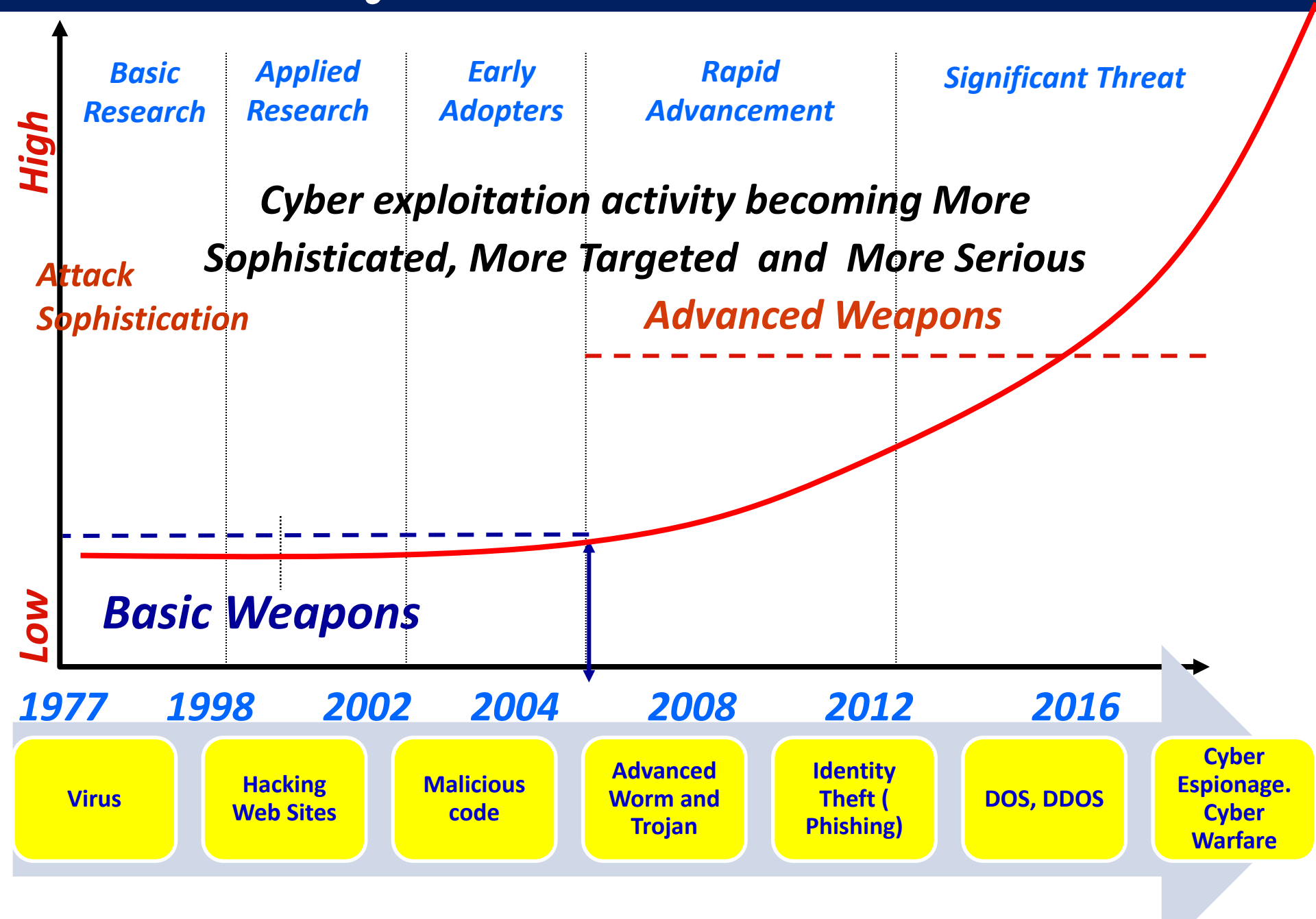
Viruses (1990s)
Anti-Virus, Firewalls

Worms (2000s)
Intrusion Detection & Prevention

Botnets (late 2000s to Current)
DLP, Application-aware Firewalls, SIM

APT, Insiders (Current)
Network Flow Analysis

Cyber Threat Evolution



Indian Cyber Situation

- India ranks 3rd in terms of the highest number of internet users in the world after USA and China, the number has grown 6-fold between 2012-2017 with a compound annual growth rate of 44%.
- India secures a spot amongst the top 10 spam-sending countries in the world alongside USA
- India was ranked among the top five countries to be affected by cybercrime, according to a 22 October report by online security firm "Symantec Corp".

Cyberattacks in India of Late

JULY 2016

UNION BANK OF INDIA HEIST

Through a phishing email sent to an employee, hackers accessed the credentials to execute a fund transfer, swindling Union Bank of India of \$171 million, Prompt action helped the bank recover almost the entire money

MAY 2017

WANNACRY RANSOMWARE

The global ransomware attack took its toll in India with several thousands computers getting locked down by ransom-seeking hackers. The attack also impacted systems belonging to the Andhra Pradesh police and state utilities of West Bengal

MAY 2017

DATA THEFT AT ZOMATO

The food tech company discovered that data, including names, email IDs and hashed passwords, of 17 million users was stolen by an 'ethical' hacker-who demanded the company must acknowledge its security vulnerabilities-and put up for sale on the Dark Web

JUNE 2017

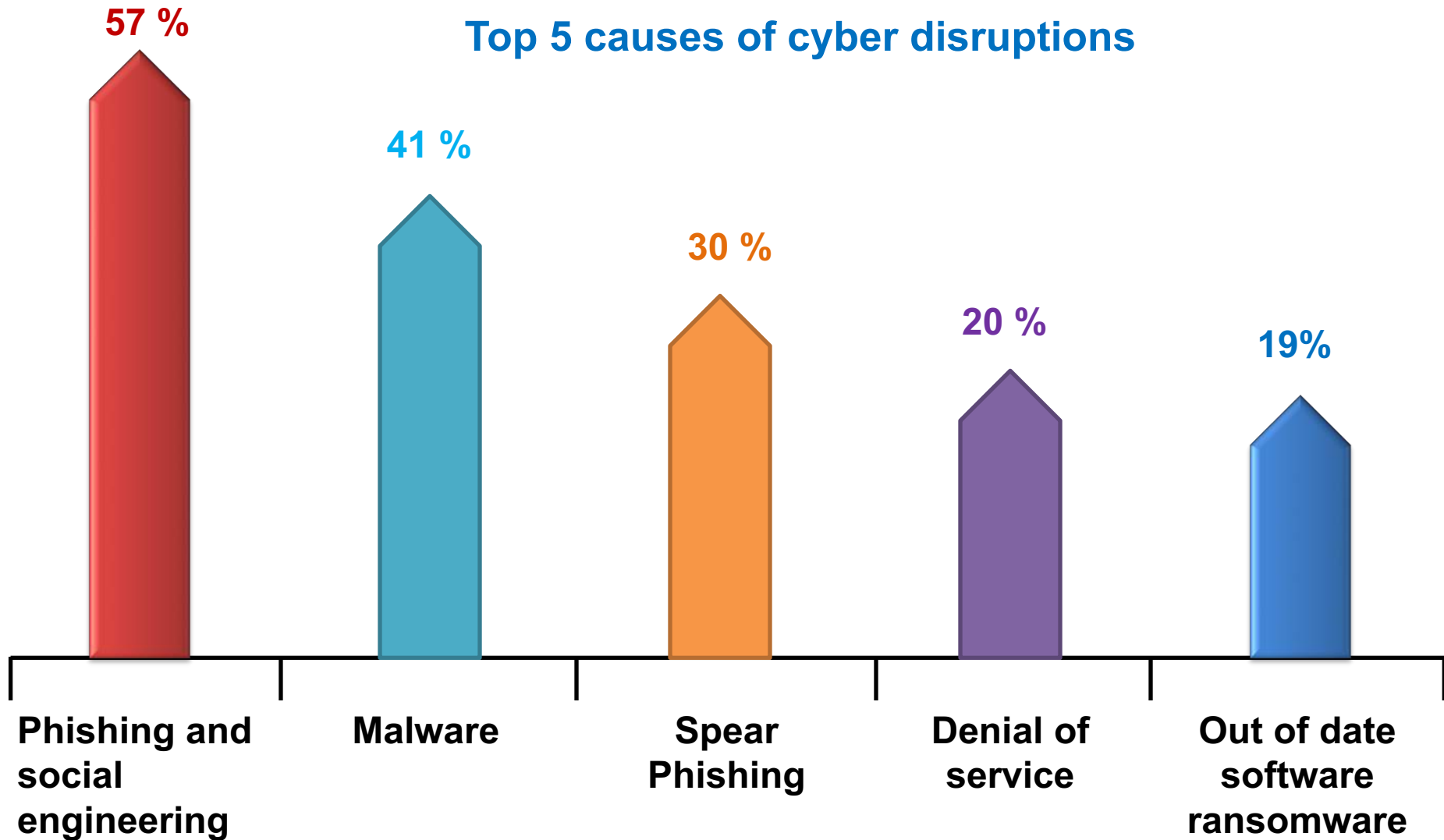
PETYA RANSOMWARE

The ransomware attack made its impact felt across the world, including India, where container handling functions at a terminal operated by the Danish firm AP Moller-Maersk at Mumbai's Jawaharlal Nehru Port Trust got affected

Cyber disruptions

> 50 % of the organizations reportedly affected in 2017

Top 5 causes of cyber disruptions



Financial and Insurance

Frequency	998 incidents, 471 with confirmed data disclosure
Top 3 patterns	1. Denial of Services, 2. Web Application Attacks and 3. Payment Card skimming Represent 88 % of all security incidents within financial services
Threat actors	94% External, 6 % Internal, <1% Partner (all incidents)
Actor Motives	96% Financials, 1% Espionage (all incidents)
Data Compromised	71% Credentials, 12 % Payment, 9% Personal
Summary	DoS attacks were the most common incident type. Confirmed data breaches were often associated with banking Trojans stealing and reusing customer passwords, along with ATM skimming operations

Source: 2017 Data Breach Investigations Report (DBIR) - Verizon

Who's behind the breaches?

75 %



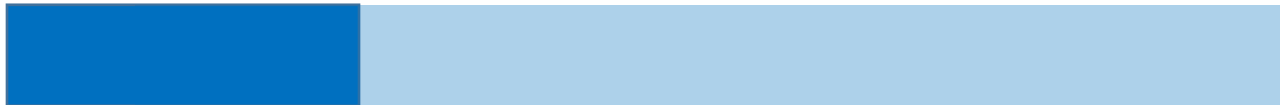
Perpetrated by outsiders .

25%



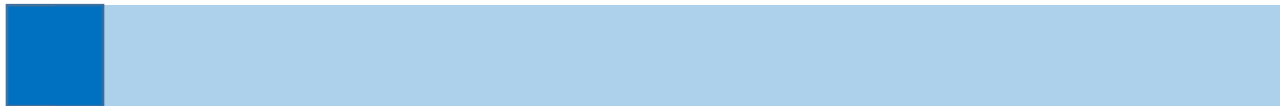
Involved internal actors.

18%



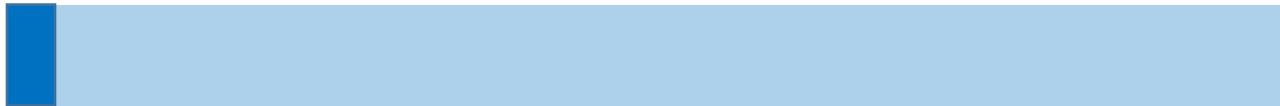
Conducted by state-affiliated actors.

3%



Featured multiple parties.

2%



Involved partners.

51%



Involved organized criminal groups.

What tactics do they use?

62 %



Of breaches featured hacking

51%



Over half of breaches included malware

81%



of hacking-related breaches leveraged either stolen and/or weak passwords.

43%



Were social attacks.

14%



Errors were causal events in 14% of breaches. The same proportion involved privilege misuse.

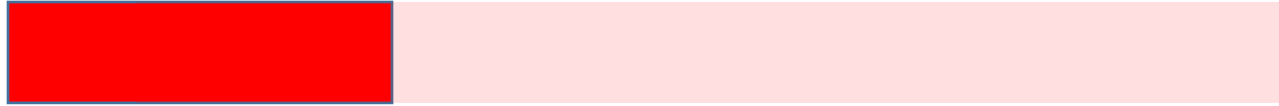
8%



Physical actions were present in 8% of breaches

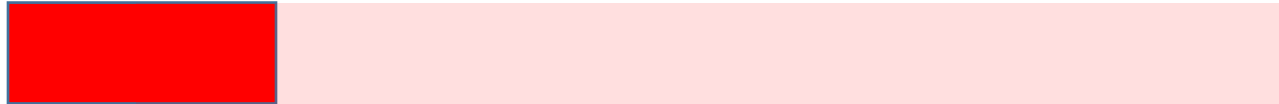
Who are the victims?

24 %



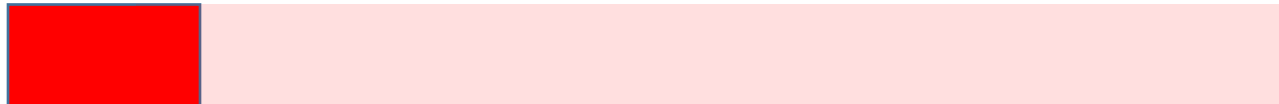
of breaches affected financial organizations.

15%



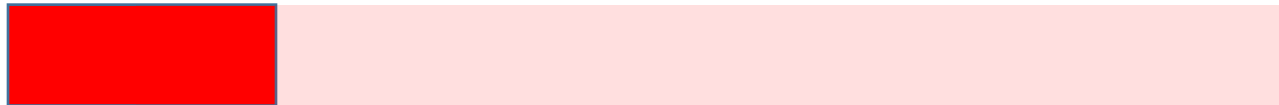
of breaches involved healthcare organizations.

12%



Public sector entities were the third most prevalent breach victim at 12%

15%



Retail and Accommodation combined to account for 15% of breaches.

What else is common?

66 %



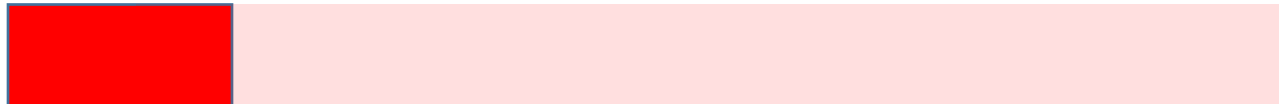
of malware was installed via malicious email attachments.

73%



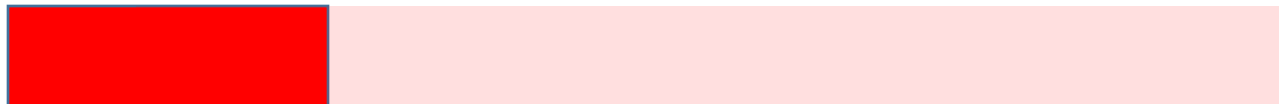
of breaches were financially motivated.

21%



of breaches were related to espionage

27%



of breaches were discovered by third parties.

International Security Trends



Cyber criminals steal personal data to affect the operations of e-commerce and finance



Organized hackers use Advanced Persistent Threat (APT) attacks to steal the confidential data of official, national defense and business



The open systems and the Internet are used in critical infrastructures increasingly. That results in the growing of the risks.



Information and security providers have been hacked to lead to damage the trusted supply chain



Cyber-warfare and DDoS paralyzed national network operations

THE BIGGEST CYBER ATTACKS

TOI



Credits: Getty Images/iStockphoto

CryptoLocker (2013)

Infected more than
2,50,000 systems

Earning **\$3 mn**

CryptoWall (2014- 16)

Extorted **\$18 million**
from victims
prompting FBI
to release an advisory

TeslaCrypt (2015)

Hit **163**
victims netting
\$76,522
for attackers

WannaCry (2017)

Hit
2,00,000+
systems



VULNERABLE TO RISKS

75%

of Indian CXOs
admit they lack
confidence in their
companies'
cybersecurity
processes

26%

Share of respondents
whose security
operations centres
collaborate and share
data with others in
the industry

69%

of respondents say their budgets to
combat cyber threats increased over
the past 12 months

CXOs surveyed: 124

Source: EY Global Information Security Survey
2016-17—India Report

RANSOMWARE STATISTICS 2017

Cybercrime
damages will cost
the world
\$6
trillion annually
by 2021

156
million phishing
emails
are sent globally
everyday

91%
of Cyberattacks
start with a
phishing
email

In 2017,
1 in 131
emails contained
Malware

According to FBI,
more than
4000
ransomware attacks
occur daily

Ransomware attacks
increased by **36%** in 2017

In 2017, every
4.2 seconds,
a new malware
specimen emerged
(according to GData)



Cybersecurity and Compliance Specialists
www.24by7security.com

Cyber Threats and Sources

Sources

- | | |
|----------------------------|---------------------------------|
| a) Nation States | b) Cyber Criminal Organisations |
| c) Terrorists, DTOs, etc., | d) Hackers / Hacktivists |

Threats

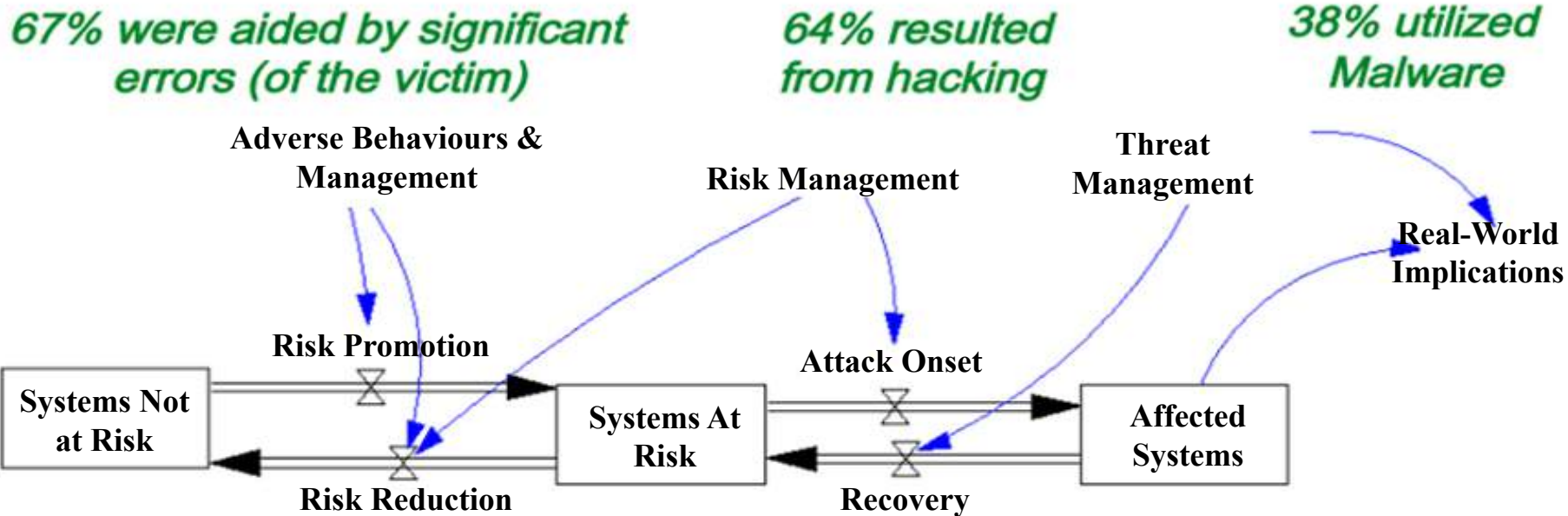
- **Malware** – Malicious software to disrupt computers
- Viruses, worms, ...
- Theft of Intellectual Property or Data
- **Hactivism** – Cyber protests that are socially or politically motivated
- Mobile Devices and applications and their associated Cyber Attacks
- **Social Engineering** – Entice Users to click on malicious links
- **Spear Phishing** – Deceptive Communications (e-mails, texts, tweets)
- Domain Name System (DNS) Attacks
- **Router Security** – Border Gateway Protocol (BGP) Hijacking
- **Denial of Service** (DoS) – blocking access to websites
- Others
- **Bottom line – easier to be a Bad Guy and volume of threats is growing**

Main *Cyber* Players and their Motives

- ***Cyber Criminals:*** Seeking commercial gain from hacking banks & financial institutions as well as phishing scams & computer ransomware
- ***Cyber Terrorists:*** Mission to penetrate & attack critical assets, and national infrastructure for aims relating to political power & “branding”
- ***Cyber Espionage:*** Using stealthy IT Malware to penetrate both corporate & military data servers in order to obtain plans & intelligence
- ***Cyber Hackivists:*** Groups such as “Anonymous” with Political Agendas that hack sites & servers to virally communicate the “message” for specific campaigns

Dynamics of Threats and Resilience (using System Dynamics modeling)

How did breaches (threats) occur?*

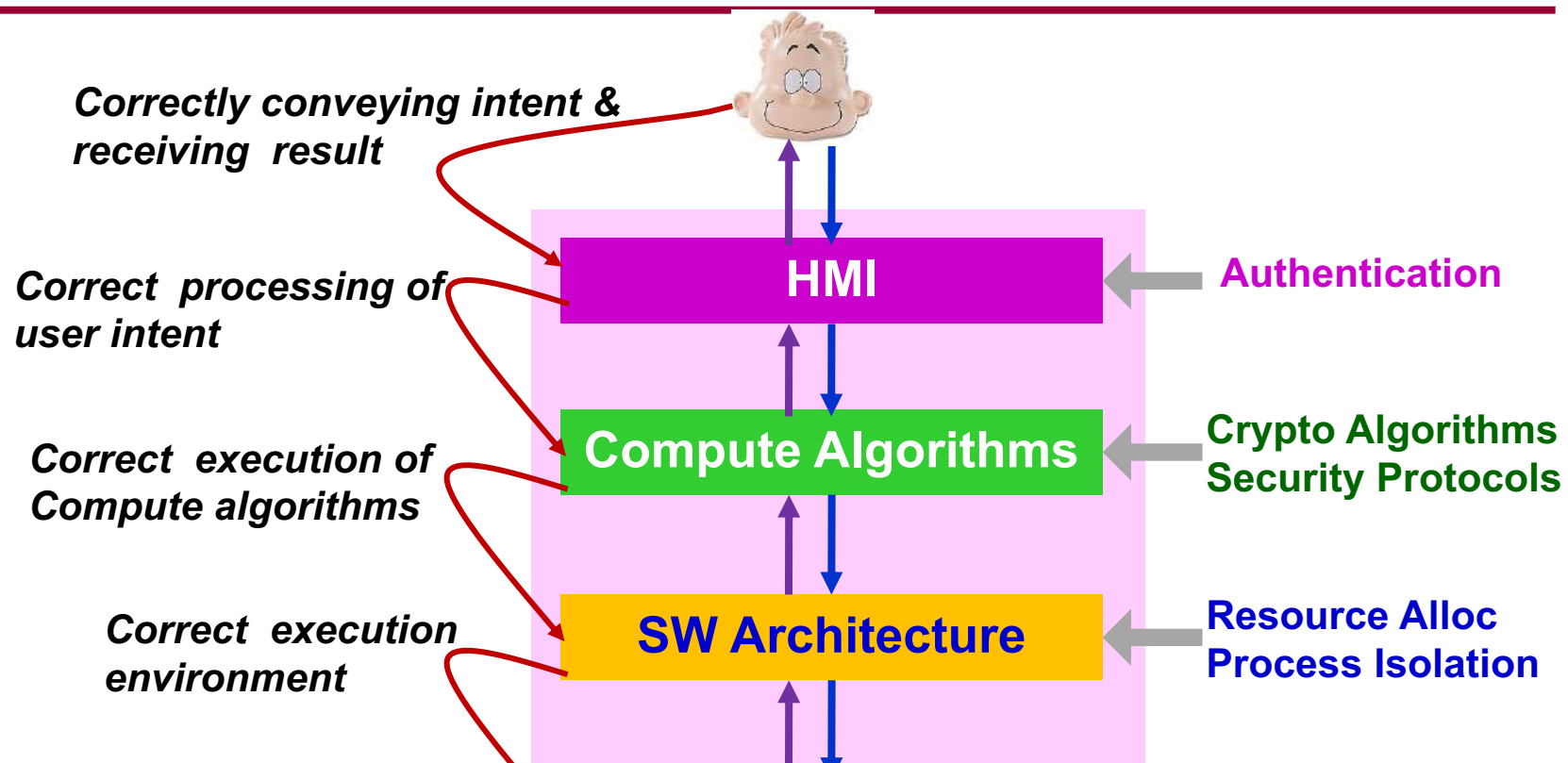


How are security and threat processes (resilience) managed? *

Over 80% of the breaches had patches available for more than 1 year

75% of cases go undiscovered or uncontained for weeks or months

Real World System



➤ Upper Layer can depend on lower layer only if it has higher

➤ Integrity

➤ Availability

➤ Software builds on hardware

➤ Hardware is the root of trust

Key Storage
Compute Logic

Security begins with a *trustworthy* hardware!!!

Hardware Cyber Security Concerns (1/2)

- ❖ Most equipment and technology for setting up Cyber Security infrastructure in India are currently procured from global sources. These systems are vulnerable to cyber threats just like any other connected system.
- ❖ There are various types of hardware attacks which includes the following.
 - Manufacturing backdoors may be created for malware or other penetrative purposes. Backdoors may be embedded in radio-frequency identification (RFID) chips and memories.
 - Unauthorized access of protected memory
 - Inclusion of faults for causing the interruption in the normal behavior of the equipment.
 - Hardware tampering by performing various invasive operations
 - Through insertion of hidden methods, the normal authentication mechanism of the systems may be bypassed.

Hardware Cyber Security Concerns (2/2)

- ❖ Above hardware attacks may pertain to various devices or systems like:
 - Network systems
 - Authentication tokens and systems
 - Banking systems
 - Surveillance systems
 - Industrial control systems
 - Communication infrastructure devices

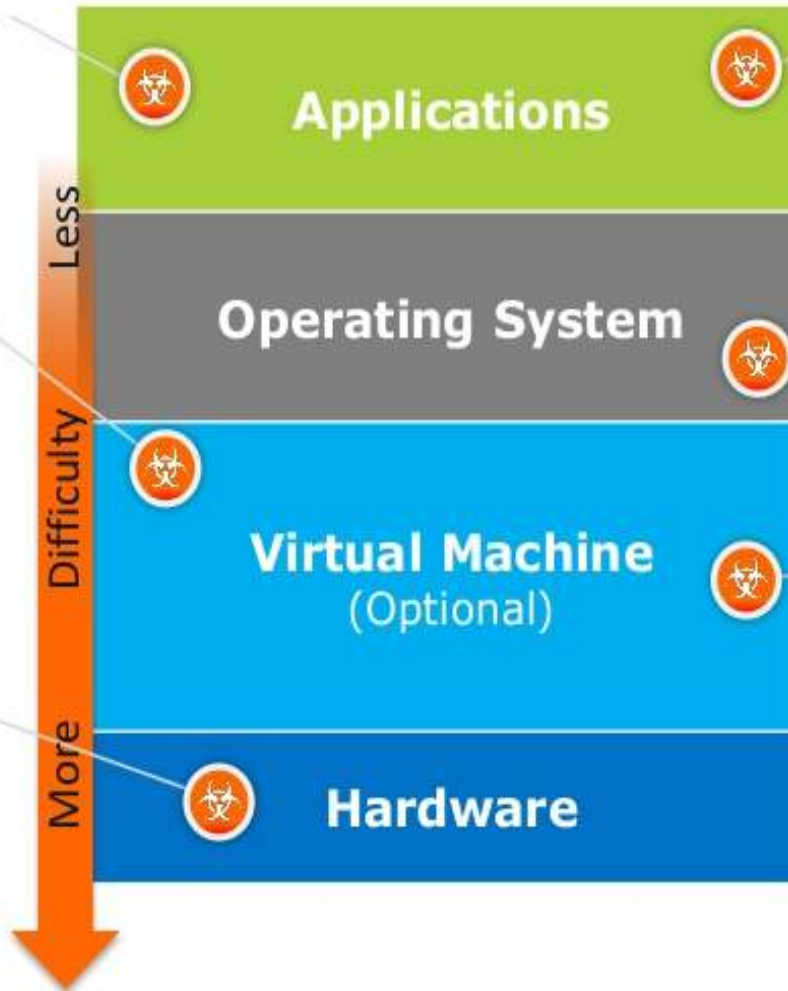
Innovations to Attack: End-Points Example

Attackers are adapting by moving down the stack:

Attacks disable security products, steal and control applications

Compromise virtual machine

Attacks against hardware and firmware affect the root-of-trust

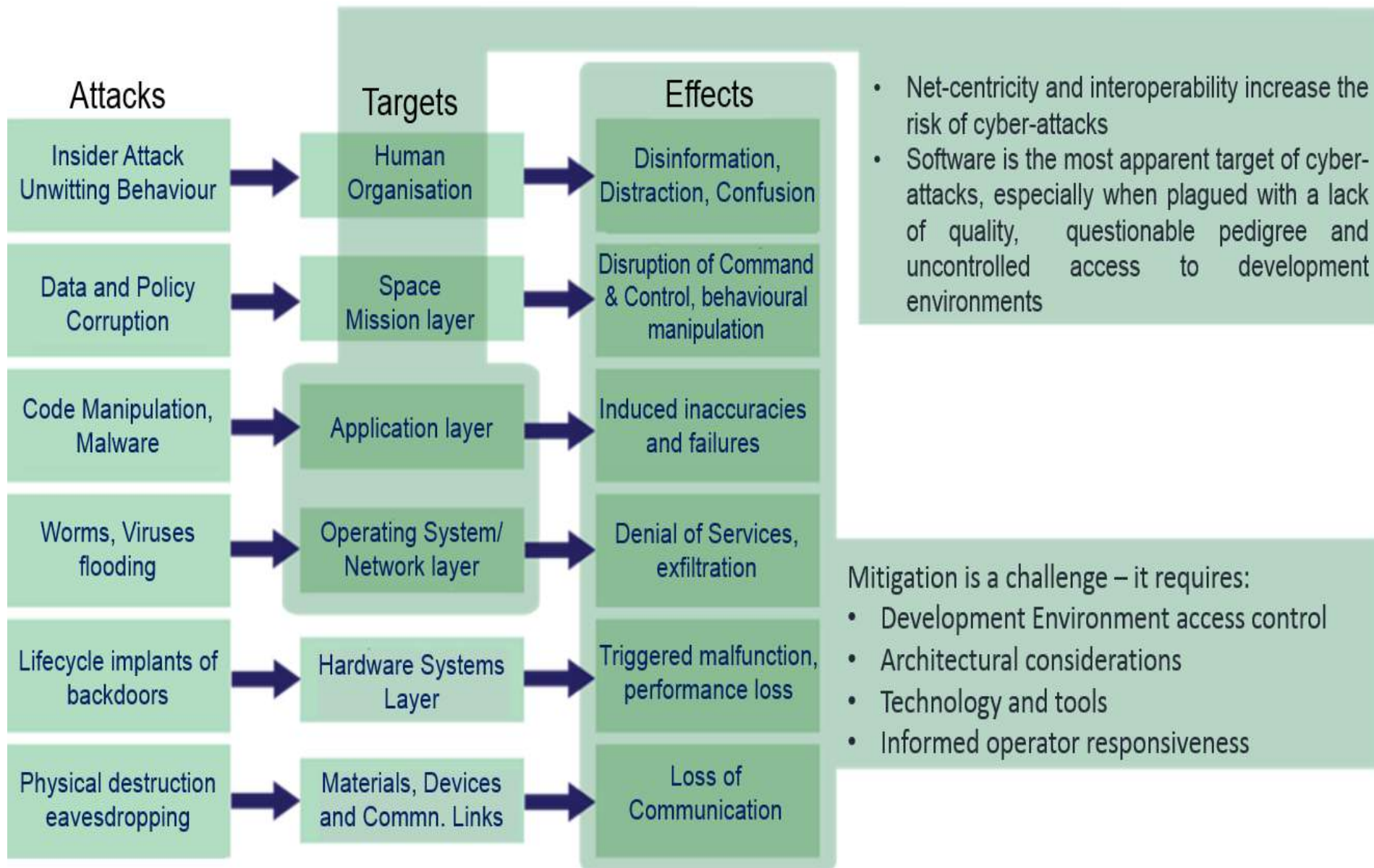


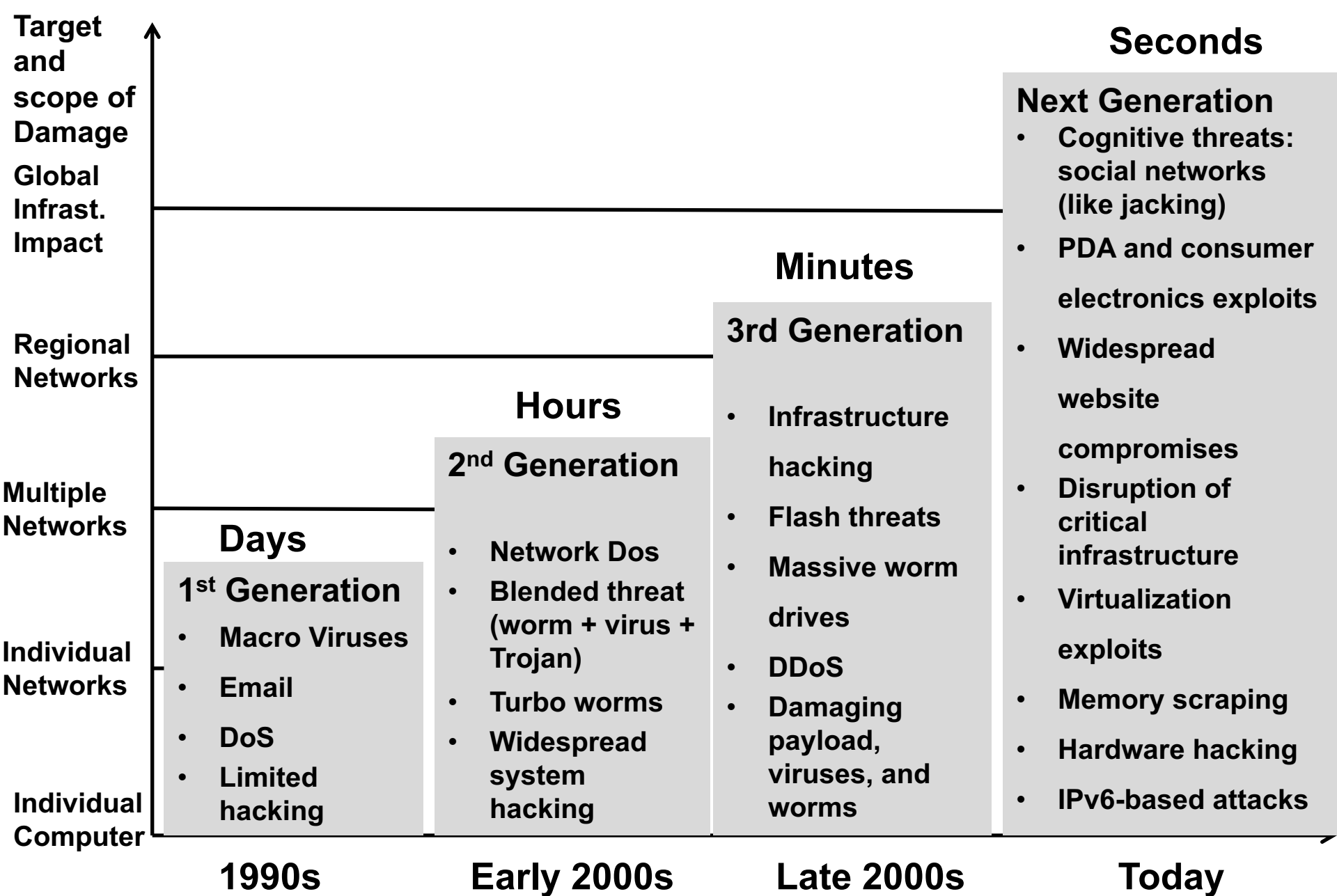
Traditional attacks:
Focused primarily on the application layer

OS infected:
Threats are hidden from security products

New stealth attacks:
Embed themselves below the OS and Virtual Machine, so they can evade current solutions

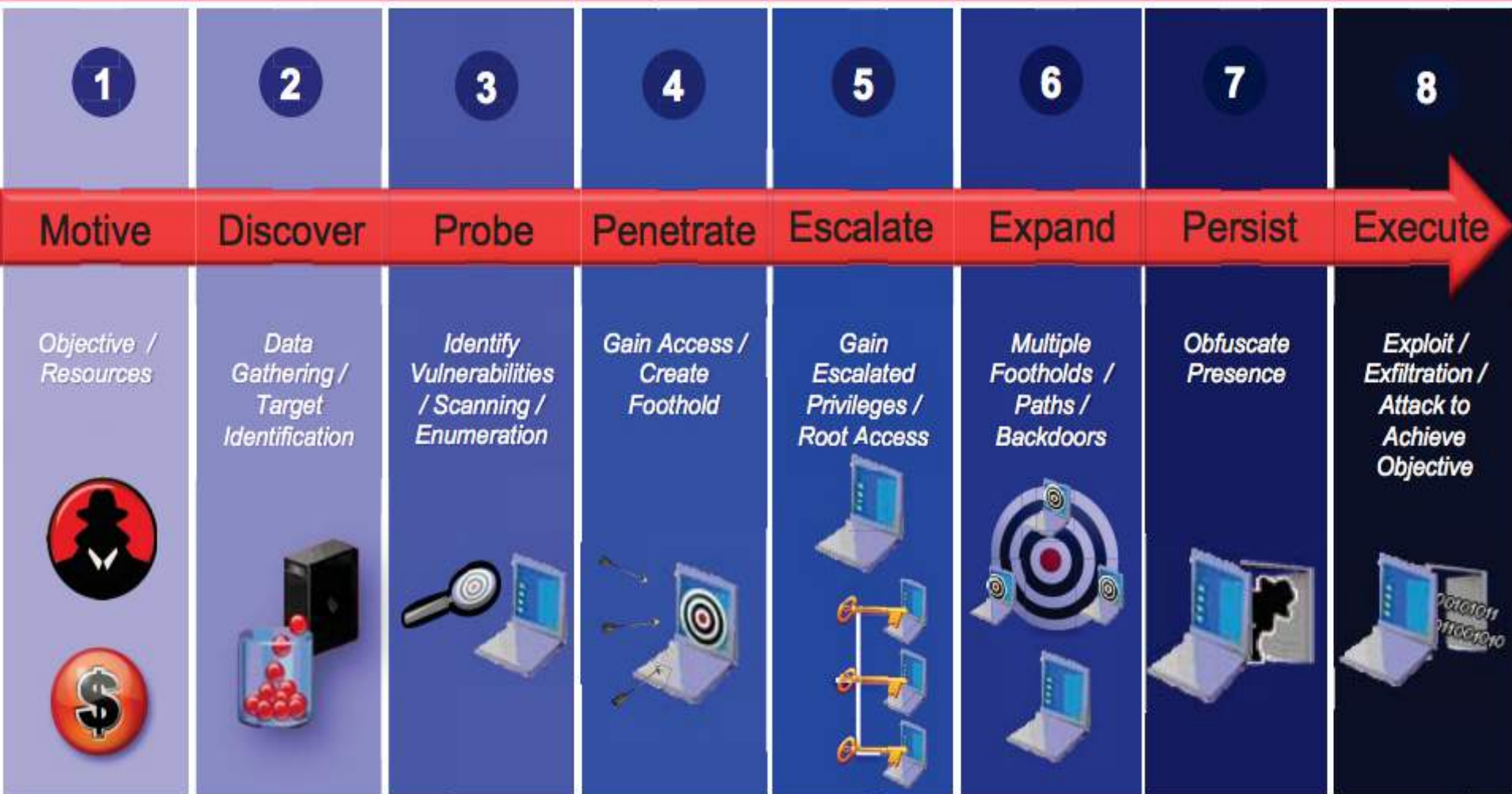
Challenges are increasing in the Cyber Space Domain





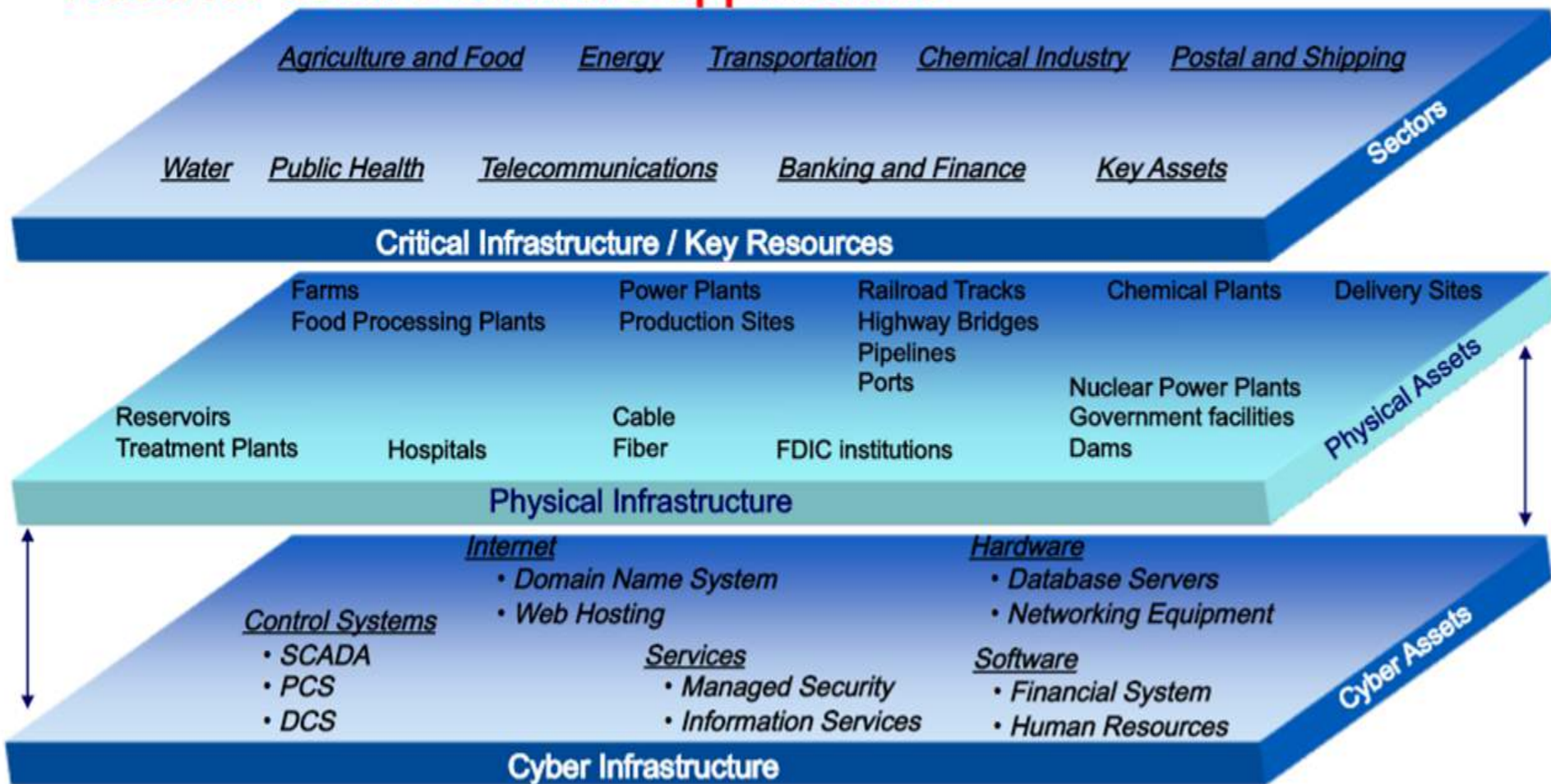
Shrinking Time Frame from Knowledge of Vulnerability to Release of Exploits

Anatomy of Attack



Cyberspace & physical space are increasingly intertwined and software controlled/enabled

Need for secure software applications



“In an era riddled with asymmetric cyber attacks, claims about system reliability, integrity and safety must also include provisions for built-in security of the enabling software.”

Cyber security

Information Security and Cryptography Technologies

System Security Technology

- Protecting Peripheral Components
- Protecting Distributed Contents
- Trusted Computing Platforms
- Detecting Intrusion/ Malware
- Protecting Data/ Access Control
- Authentication

Network Security Technology

- Protecting Privacy or Anonymity
- IEEE 802.11
- Security Pairing

Wireless Network Security Technology

- Security at particular protocol layers
- Detecting Malicious traffic
- Key Management

Cryptography

- Symmetric Cypher
- Asymmetric Cypher
- PKI-Digital certificate
- Secure hashing
- Key Management
- Quantum Cryptography

Cyber Security Framework

User

Assets

Transactions

Governance

Identification and
Authorisation

Privacy
Minimum Disclosure
Anonymity Support

Data Security

Sovereignty
Data Localisation
Interoperability
Secure Communication

Threat
Management

Profiling
Protection
Detection
Response

Building Resilience

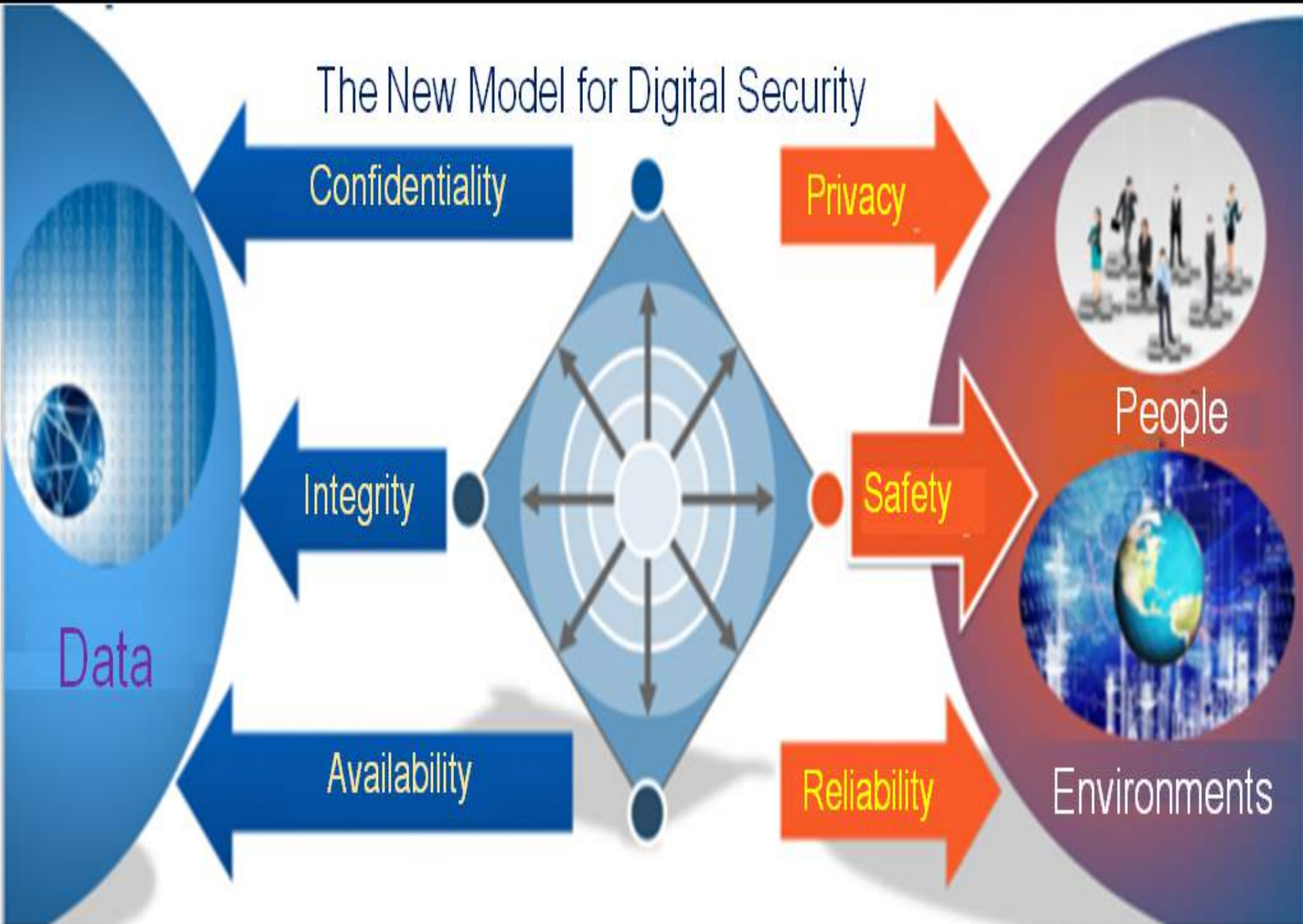
Risk based decisions
Across Data Flow
People Centric Security

Visibility

Analytics

Integration

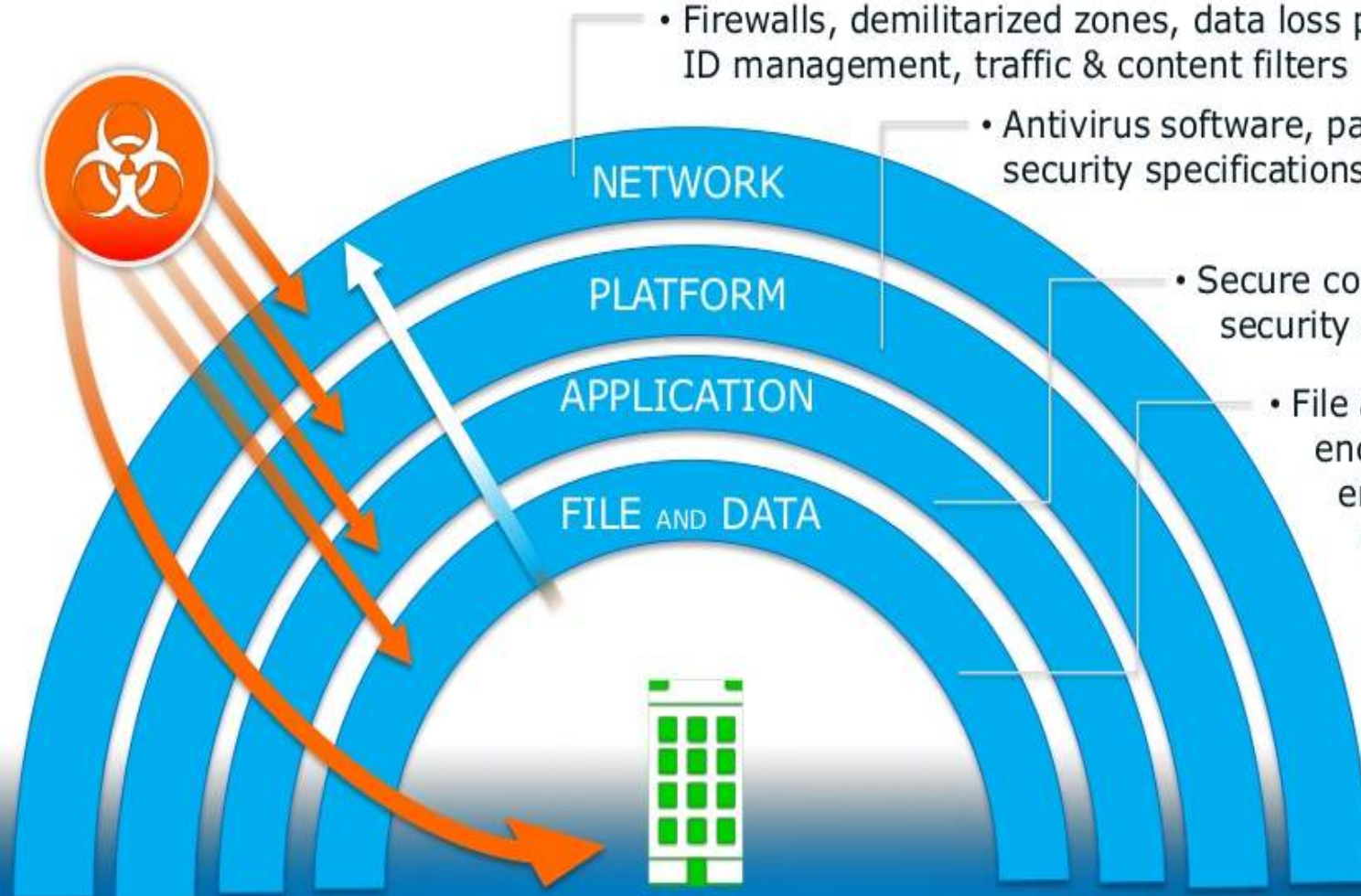
The New Model for Digital Security



Tactical Security Technology Integration: Layered Defense

Multiple layers are necessary for comprehensiveness

- Firewalls, demilitarized zones, data loss prevention, ID management, traffic & content filters
- Antivirus software, patching, minimum security specifications for systems
- Secure coding, testing, security specifications
- File and data encryption, enterprise rights management



IoT Cyber Security - Vulnerabilities

Operational Security

IOT Bases services require continuity and high availability

Privacy

Valuable Data Require Protection

Software Patching

Many IoT devices lack human users who can install security updates

Identity of Things

In the absence of universal standards, each implementation requires unique approach to manage authentication and access

Logging

Logging System must identify events without relying time of day data

Future Technology to be Designed with Security

Security must be part of the design for future technology. Adding security after, is no longer sufficient or sustainable



Smart

Security innovation must deliver more capable solutions to keep pace with threats



Open

Platforms and security standards must be open to promote collaboration and accelerate adoption



Trusted

Technology and security providers must be trustworthy in the creation and operation of their products



Strong

Products and services must be hardened to resist compromise and make security transparent to users

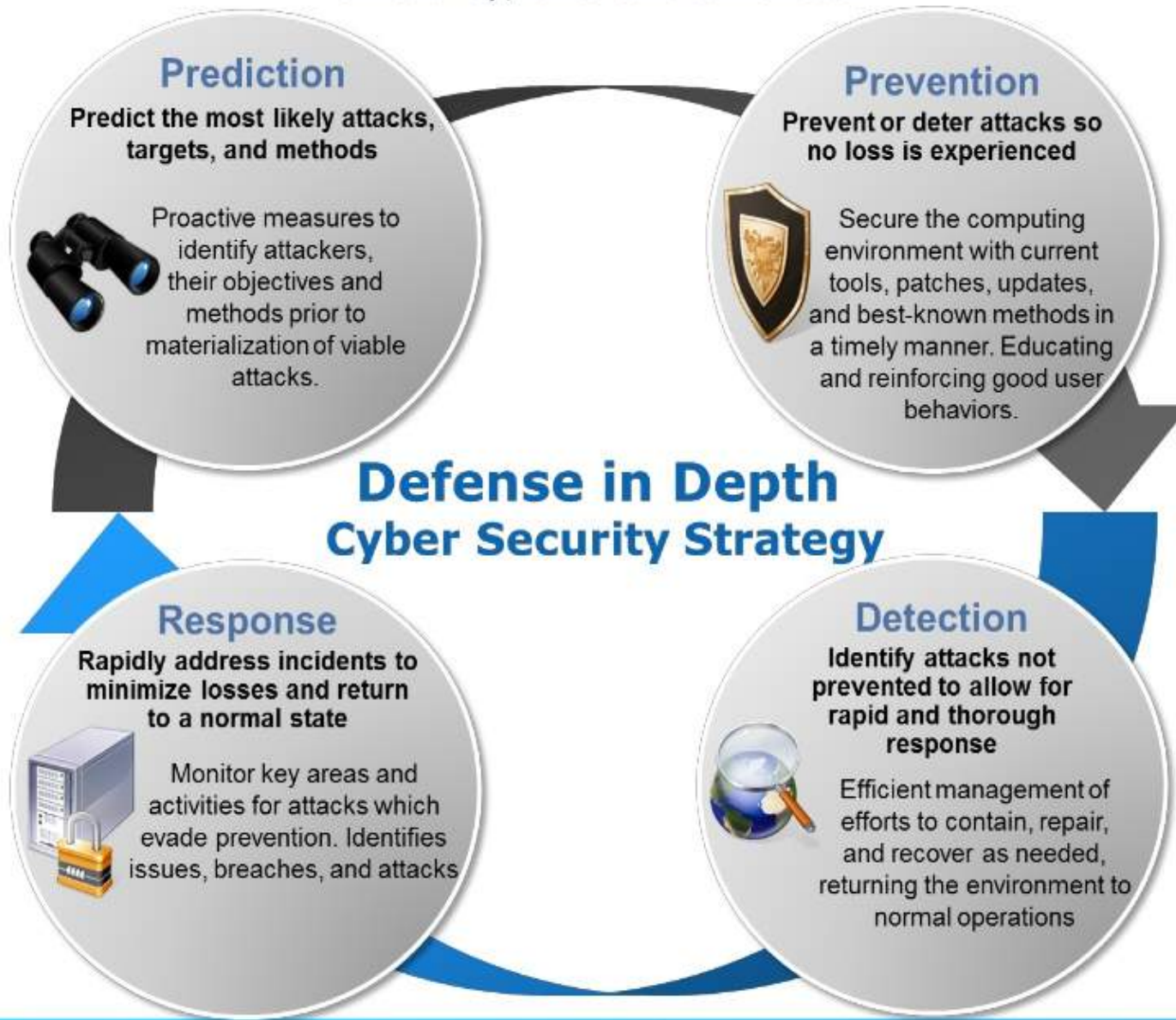


Ubiquitous

Security must protect data wherever it exists or is used, for all parties and devices across the compute landscape

Strategic Leadership: Defense in Depth

A strong process strategy will enable operational flexibility, while driving cost efficiency, and effectiveness



10 Steps to Cyber Security (1/5)

Defining and communicating your Board's Information Risk Management Regime is central to your organisation's overall cyber security. CESG recommend you review this regime – together with the nine associated security area described below – in order to protect your business against the majority of cyber threats

1. **Network Security**

Protect your networks against external and internal attack. Manage the network perimeter. Filter out unauthorised access and malicious contents. Monitor and test security controls

2. **Malware Protection**

Produce relevant policy and establish anti-malware defences that are applicable and relevant to all business areas. Scan for malware across the Orgn.

10 Steps to Cyber Security (2/5)

3. **Monitoring**

Establish a monitoring strategy and produce supporting policies. Continuously monitor all ICT system and networks. Analyse logs for unusual activity that could indicate an attack.

4. **Incident Management**

Establish an incident response and disaster recover capability. Produce and test incident management plans. Provide specialist training to the incident management team. Report criminal incidents to law enforcement.

10 Steps to Cyber Security (3/5)

5. **User Education and Awareness**

Produce user policies covering acceptable and secure use of the organisation's systems. Establish a staff training programme. Maintain user awareness of the cyber risks.

6. **Home and Mobile Working**

Develop a mobile working policy and train staff to adhere to it. Apply the secure baseline to all devices. Protect data both in transit and at rest

10 Steps to Cyber Security (4/5)

7. **Secure Configuration**

Apply security patches and ensure that the secure configuration of all ICT systems is maintained. Create a system inventory & define a base line build for all ICT devices.

8. **Removable Media Controls**

Produce a policy to control all access to removable media. Limit media types and use. Scan all media for malware before imported on the corporate system.

10 Steps to Cyber Security (5/5)

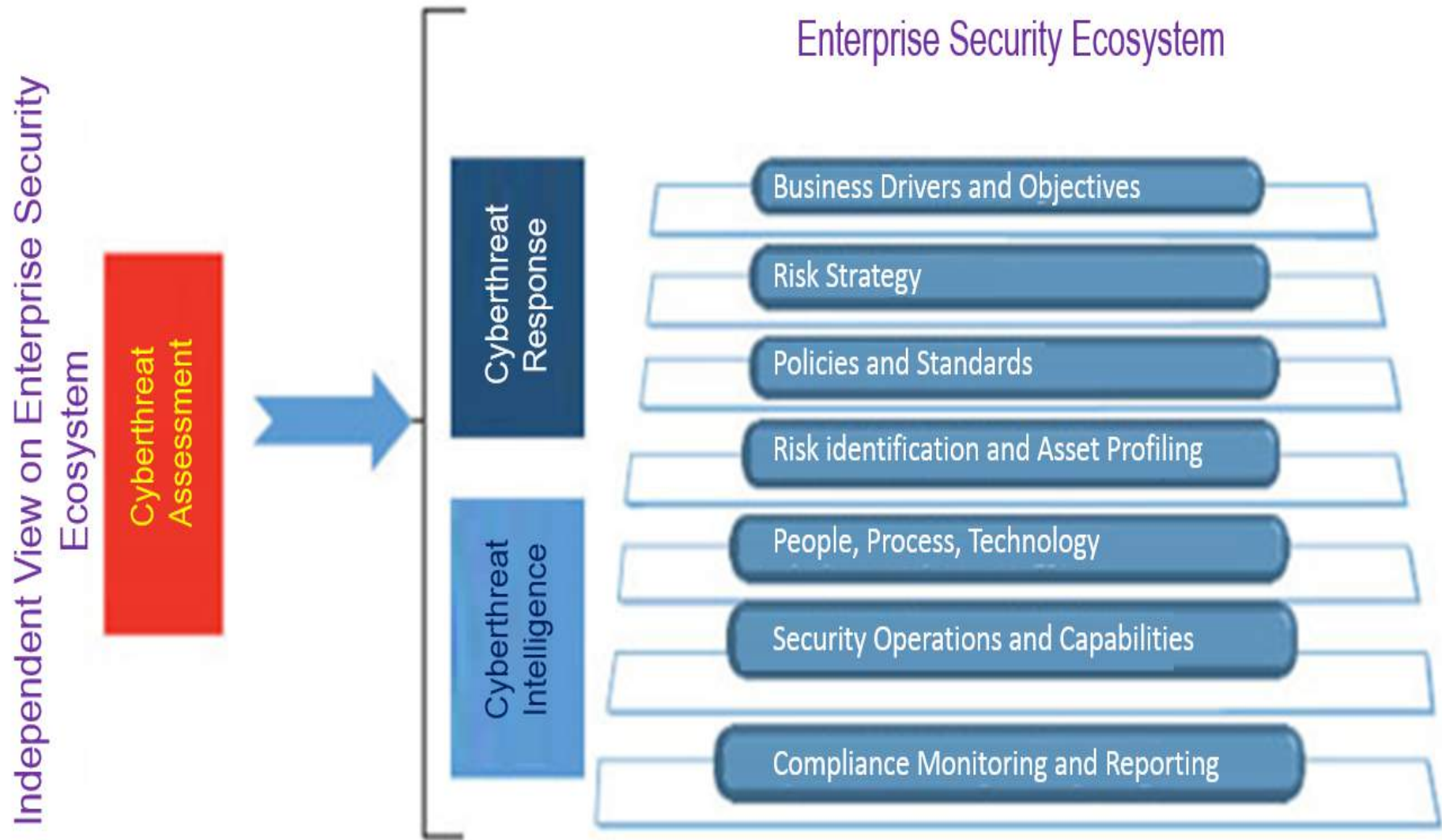
9. **Managing User Privileges**

Establish account management processes and limit the number of privileged accounts. Limit user privileges and monitor user activity. Control access to activity and audit logs.

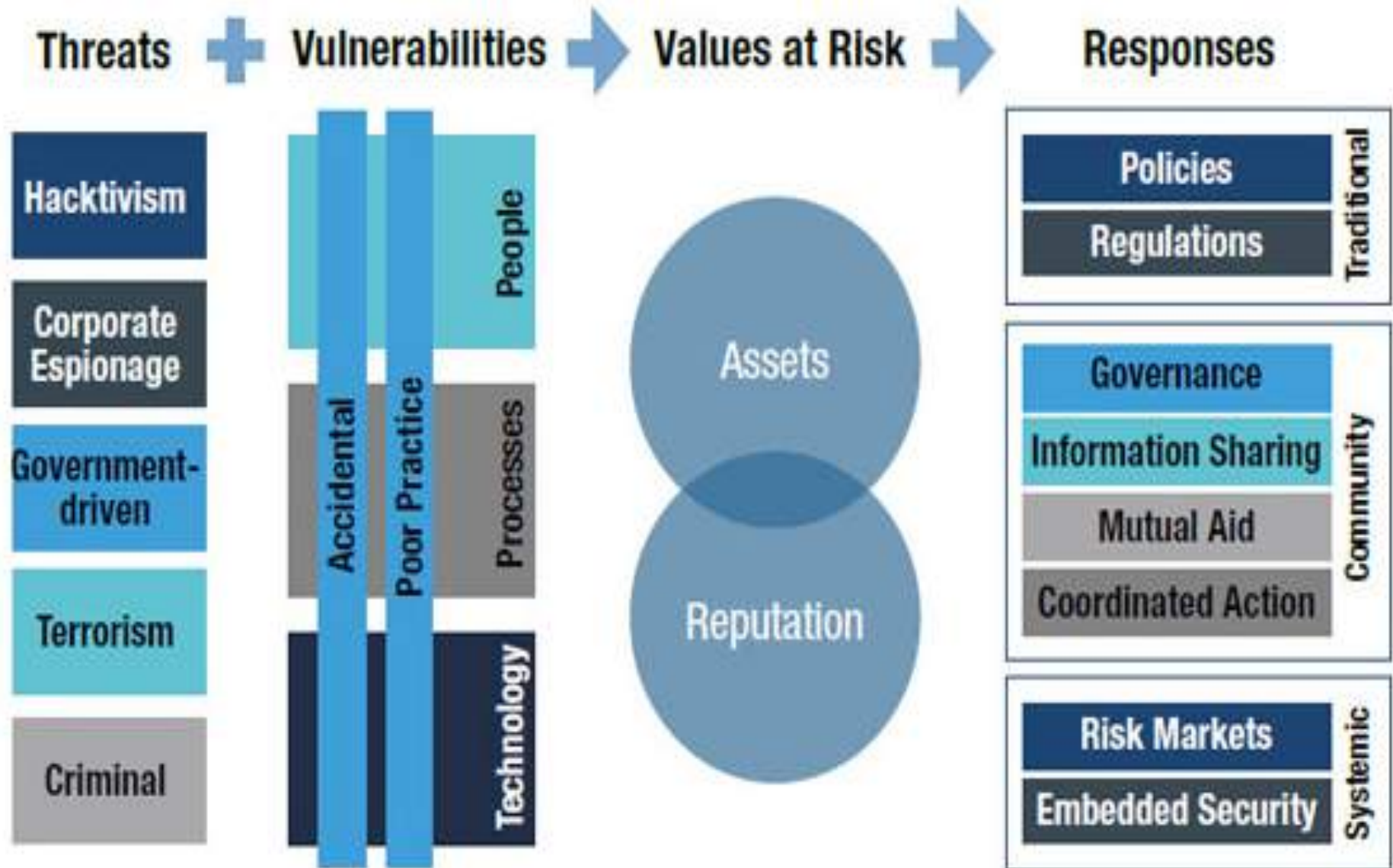
10. **Information Risk Management Regime**

Establish an effective governance structure and determine your risk appetite. Maintain board engagement with cyber risk. Produce supporting information risk management policies.

Evolution of Cyberthreat Management



World Economic Forum Cyber Risk Framework



Cyber Security Planning

Cyber Trust Layer (1)

Organization heads meet and discuss the criteria for their mutual cyber trust. Network vulnerabilities will be ranked by this criteria

Cyber Impact Layer (2)

Organization heads meet and discuss how their mutually agreed upon criteria will affect their cyber risk. The result drives criteria weights.

Cyber Analytic Layer (3)

Cyber security researchers develop a set of custom scientific metrics to assess the Global Severity of identified network vulnerabilities which are ranked by the agreed upon criteria

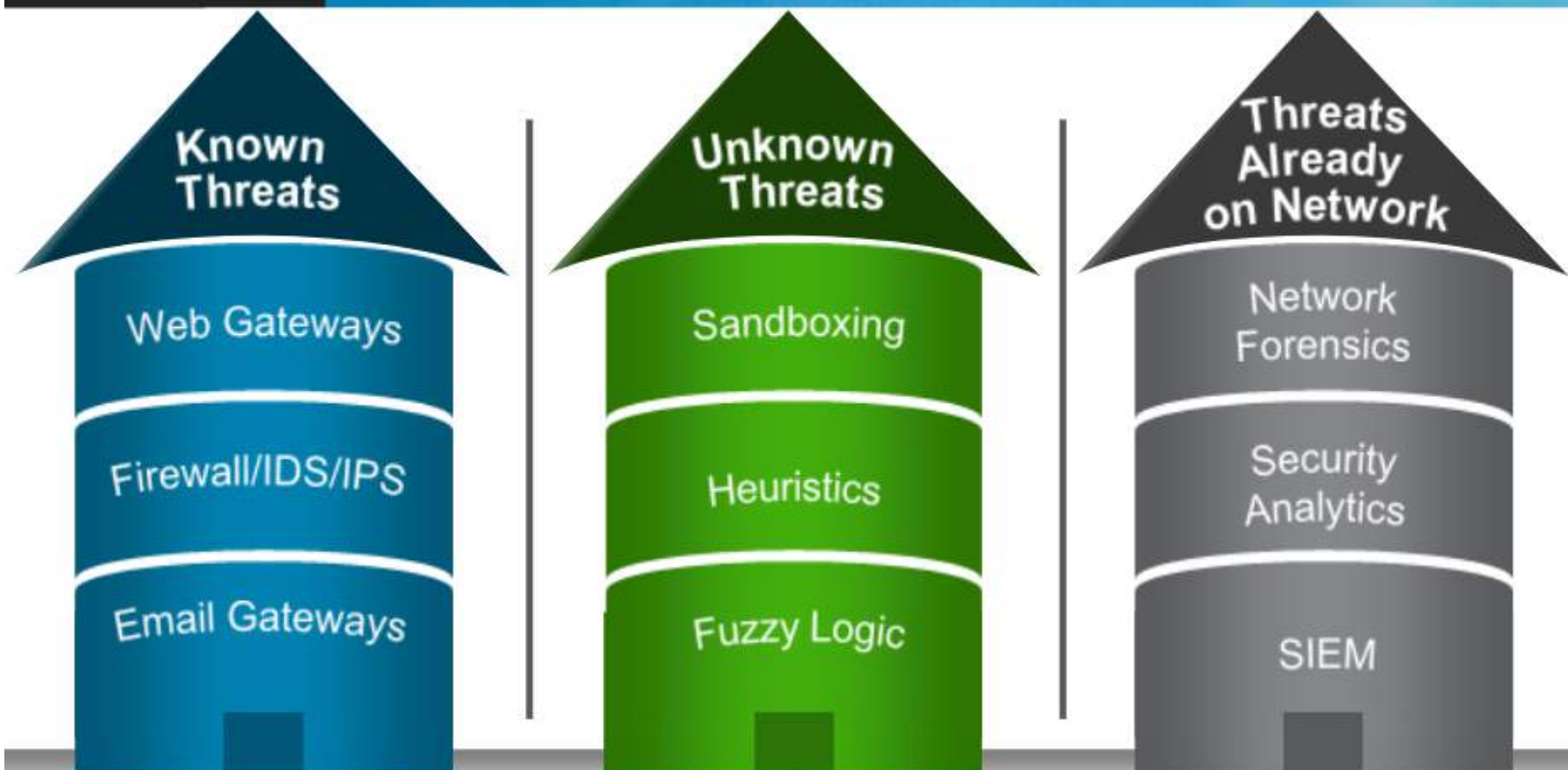
Cyber Metric Implementation Layer (4)

Cyber security researchers meet with network engineers to develop continuous monitoring framework to identify vulnerabilities and to calculate metrics.

Cyber Evaluation Layer (5)

Organization heads meet to discuss their individual Cyber Maturity and to determine the minimum risk-level necessary for collaboration

CURRENT SOLUTIONS OPERATE IN SILOS



Technology and Organizational silos limit current defenses



INTELLIGENT DEFENSE IN DEPTH



Technology Shifts and Disruptions

Digital
Technologies

Cloud
Computing

Automation &
Robotics

Artificial
Intelligence
& Cognitive
Systems

Internet
of
Things

Virtual
Reality

Scope and Overview

Tech
Shifts

- Technology shifts and trends that are going to disrupt current business models and industries in next 4-5 years; and the impact therefore on Business and Indian IT Industry and how do we take advantage of it.

Enabling Technologies

Industry & Business Themes of Digital Economy

Impact

Analytics/Big Data

Mobility/Mobile Internet

Cloud

Artificial Intelligence

Connected world (IoT)

Automation & Robotics

Augmented & Virtual
Reality

Others?

- Finance Services
- Communication
- Media & Entertainment
- High Technology
- Healthcare
- Public service
- Natural Resources
- Retail & CPG
- Travel & Hospitality
- Infrastructure
- Manufacturing & Ind Equipment
- Energy & Utility
- Automotive
- Education

Seamless Payment

Connected Healthcare

Sustainable
Manufacturing

Immersive Learning

Painless Travelling

Personalized
Consumptions

Everything as a Service

Cybersecurity Market Sectors

- Anti-Virus/Firewall
- ID Authentication
- Encryption/Privacy
- Risk & Compliance
- Mobile Device Security
- Anti-Fraud Monitoring
- Website Protection
- S/W Code Verification
- AI & Machine Learning
- Enterprise IoT Security
- Cloud Security Services
- Big Data Protection
- RT Log/Event Analytics
- Real-Time Threat Maps
- Smart Biometrics
- Training & Certification

Global Trend is towards ***Adaptive & Intelligent Cybersecurity Solutions/Services...***
....Traditional ***Anti-Virus/Firewall Tools*** no longer fully effective against ***"Bad Guys"***!

Trends, Challenges and Threats in 2018 (1/5)

1. AI and machine learning can boost cyber defenses

- As artificial intelligence and machine learning gathers pace, and starts to impact more and more industries, it's sure to play a bigger role in cybersecurity.
- Because the battle with cyber criminals moves so quickly, machine learning models that can predict and accurately identify attacks swiftly could be a real boon for InfoSec professionals.
- These models need to be trained and honed. However, there is also a risk that AI and machine learning may be exploited by attackers.

Trends, Challenges and Threats in 2018 (2/5)

2. Be proactive about ransomware

- Ransomware has been a growing threat for the last few years, but it continues to claim high profile victims.
- It's not yet clear what everyone learned from the WannaCry Ransomware attacks, highlighted the need to back up regularly, keep patching and updating systems, and strengthen your real-time defenses. If organizations took these simple steps, we could dramatically reduce the impact of ransomware.

3. Handling data breaches gracefully

- It may prove impossible to eradicate data breaches completely, but every organization has the power to lessen the blow by handling the aftermath correctly.
- Equifax gave us a masterclass in how not to handle a data breach earlier this year. By delaying disclosure, misdirecting potential victims, and failing to patch a known vulnerability, one can make a instructive for others in the year ahead.

Trends, Challenges and Threats in 2018 (3/5)

4. The IoT is a weak link

- We're rolling out more and more sensor-packed, internet-connected devices, but the Internet of Things remains a major weak point for defenses.
- All too often these devices lack basic security features, or they aren't properly configured and rely upon default passwords that can give attackers easy access.
- This in turn is giving rise to botnets, which can be used for volumetric attacks, to exfiltrate stolen data, to identify further vulnerabilities, or for brute force attacks. We need to properly secure the IoT or it will continue to be a big issue in 2018.

Trends, Challenges and Threats in 2018 (4/5)

5. There's still a skills shortage

- The dearth of skilled cybersecurity professionals continues to be a major problem for many organizations.
- Even with average InfoSec salaries soaring, there are thousands of vacant positions.
- This is leading many companies to engage external cybersecurity services and virtual CISOs. We expect to see more outsourcing as employers try to find a way to fill the skills gap.

6. Developing a common language

- While the specter of multiple threats looms, there are also positive developments in the cybersecurity realm, not least the creation and adoption of things like NIST's Cybersecurity Framework.
- As more organizations and cybersecurity experts come together to develop a common language, our collective defenses grow stronger.

Trends, Challenges and Threats in 2018 (5/5)

7. Patching and application testing

- It's not shiny or new or exciting, but it should still be top of mind. The number of data breaches in 2017 that were made possible by known vulnerabilities and a sluggish approach to patching is horrifying. It's not enough to identify problems – you must act.
- Application testing falls into the same bucket, in that it's too often ignored.
- If you don't test your security, then you don't know how secure your application is.
- If everyone put a fresh effort into patching and app testing in the coming year, we would see a dramatic drop in data breaches.

Cyber Security HR Requirements

- **Challenge**
 - Acute Shortage of Resource persons
 - Inadequate research in academia
 - **Trustworthy System Design: Multidisciplinary Field**
 1. Computer Science
 2. Electronics and Computational System Engineering
 3. Software Engineering
 4. Information Technology
 - **Such courses currently not offered in India**
 - Courses can be developed
 - Offered over NKN in MOOC model
-

Human Resource Development

- **Specialists in Trustworthy Information Systems Engineering**
- **Build Curriculum at UG/PG/PhD Levels**
- **Courses should be offered in three tracks**
 - **Systems Area**
 - Focus on Attacks from within the system boundary with an emphasis on platform, operating systems, and secure system development.
 - **Networks Area**
 - Focus on protecting information assets from network-based intrusion and from attacks that are primarily focused on remote exploitation of protected systems.
 - Cybersecurity approaches that are effective in this paradigm should be explored in depth and various defensive approaches should be investigated.
 - **Analysis Area**
 - Focus on both the systems and networks tracks. Analysis courses study low-level behavior, code, and data to understand anomalies and develop the ability to identify unexpected patterns and malicious events.

The Future of Cyber security Education is Bright

- Emerging challenges will drive the needs in cybersecurity – *Understand the market needs*
- Employers will expect workers to know and apply industry best practices and perspectives - *Align academics to the future expectations*
- The roles are expanding for incoming cybersecurity workforce - *Prepare students for the new roles*
- Resources are emerging to assist academic staff and graduates to understand the needed skill and opportunities – *Empowered students to be self-sufficient in tracking employment demands*



➤ Foundations

- Security Models
- Formal methods
- Cryptography

➤ Application-Centric

- Secure information sharing
- Social computing
- Health care
- Data provenance

➤ Technology-Centric

- Cloud computing
- Smart grid
- Trusted computing

➤ Attack-Centric

- Botnet and malware analysis
- Complex systems modeling
- Zero-day defense
- Moving target defense

CYBER SECURITY HIERARCHY IN INDIA (1/2)					
PM OFFICE/CAB INET SECY (PMO/CAB SEC)	MINISTRY OF HOME AFFAIRS (MHA)	MINISTRY OF EXTERNAL AFFAIRS (MEA)	MINISTRY OF DEFENCE (MOD)	MINISTRY OF COMMON INFO TECHNOLOGY (MCIT)	NON GOVT ORGANIZATION (NGO)
National Security C0uncil (NSC)	National Cyber Corrd Centre (NCCC)	Ambassadors & Ministers	Tri Service Cyber Commad	Department Of Information Technology (DIT)	Cyber Security And Anti Hacking Organisation (CSAHO)
National Technical Research Org (NTRO)	Directorate of Forensic Science (DFS)	Defence Attaches	Army (MI)	Department of Telecom (DoT)	Cyber Society of India (CySI)
National Critical Info Infrastructure Protection Centre (NCIIPC)	National Disaster Mgt Authority (NDMA)	Joint Secretary (IT)	Navy (NI)	Indian Computer Emergency Response Team CERT- IN	Centre of Excellence for Cyber Security Research & Development in India (CECSRDI)
Joint Intelligence	Central Forensic Science Lab (CFSLs)		Air Force (AFI)	Educational Research Network (ERNET)	Cyber Security of India (CSI)

CYBER SECURITY HIERARCHY IN INDIA (2/2)					
PM OFFICE/CABINET SECY (PMO/CAB SEC)	MINISTRY OF HOME AFFAIRS (MHA)	MINISTRY OF EXTERNAL AFFAIRS (MEA)	MINISTRY OF DEFENCE (MOD)	MINISTRY OF COMMON INFO TECHNOLOGY (MCIT)	NON GOVT ORGANIZATION (NGO)
National Crisis Management Committee (NCMC)	Intelligen ce Bureau (IB)		Def Info Assurance & Research Agency (DIARA)	Informatics Center (NIC)	National Cyber Security of India (NCS)
Research & Analysis Wing (RAW)			Defence Intelligence Agency (DIA)	Centre for Development of Advanced Computing C-DAC	Cyber Attacks Crisis Management Plan of India (CACMP)
Multi Agency Center			Defence Research Dev Authority (DRDO)	Standardisati on, Testing and Quality Certification (STQC)	
National Information Board (NIB)					

Recommendations on Cybersecurity Framework for States (1/4)

P-P-P Model for Cybersecurity

- State Cybersecurity Framework shall be envisaged in P-P-P Model
- Government shall partner with the private sector and the academia to strengthen cybersecurity posture of the state

Information Security Policy and Practices

- IS Policies & practices shall be mandated at govt. functionaries & its service providers
- Security Audit Adhering to international standards applicable for all govt. websites, applications before hosting and publishing
- Govt. to ensure ISPs operating in the state shall deploy cybersecurity plans in line with State cybersecurity policy

Recommendations on Cybersecurity Framework for States (2/4)

State Computer Emergency Response Team

- Establishment of the State CERT to operate in conjunction I-CERT and coordinate with NCIIPC
- Cybersecurity drills shall be carried out under the supervision of I-CERT

Identity Theft and Security Incident Prevention

- State cybersecurity framework to support strategy and implementation mechanisms to prevent digital impersonation and identity theft and the security incidents

Recommendations on Cybersecurity Framework for States (3/4)

Assurance Framework

- Framework of assurance shall be established to provide guidance on security certifications, qualification criteria and prescribe security audits of gov. ICT systems, Projects & applications

Security Budget

- Govt. agencies implementing IT Projects shall allocate appropriate budget towards compliance with the security requirement of IT Act 2000 and State cybersecurity policy, ISMS, security solution procurement and trainings.

Recommendations on Cybersecurity Framework for States (4/4)

Information Sharing

- State Information Sharing Network for CII shall be established

Capacity Building and Awareness

- Govt. shall take appropriate steps for enhancing awareness of citizens and small business for cybersecurity
- Cybersecurity Capacity building and training for professional, extending ISEA program, introducing curricula academia and organizing conferences
- Strengthening LEAs through training, establishment of forensics labs, etc.

Towards 2025 : ***“Smart Security Solutions”***

- The Application of Artificial Intelligence and Machine Learning allows us to develop ***“Smart Security Solutions”*** as follows:

.....***“Smart Security Solutions”*** typically possess the following features:

- 1) ***Space-Time Awareness:*** Location (GPS) & Real-Time Clocks
- 2) ***Learning, Adaptation & Self-Organisation:*** Real-Time Intelligence
- 3) ***Massive Memory & Storage:*** Local & Remote Cloud Storage
- 4) ***Sustainability:*** Embedded Security – *Everywhere in the Network!*
- 5) ***Scalable Networked Architecture:*** Smart Architectures will need to scale in space & time from micro cells to macro solutions
- 6) ***Decision Focus:*** “Knowledge Lens” for Data Mining & “Big Data” from Global Social Networks, Search & On-Line Trade & Commerce
- 7) ***Systems Integration:*** Cyber and Physical Solutions & Operations



THANK YOU